

QUIZ - 1A
STATISTICAL COMPUTING (MTH-210)

Name:

Roll Number

TIME 20 MINS.

MAXIMUM SCORE 15 MARKS

Please write your name and roll number, otherwise two marks will be deducted. This quiz has two questions and each question has two subparts. You have to write the correct answer only in the appropriate place. Keep all your codes with you till the end of this course. I might ask for it for cross verification. If you cannot provide the code then you will be assigned zero for that particular question. We use the notations $X_{(1)} < \dots < X_{(n)}$ as the ordered values of $X_1, \dots, X_n$.

1. Generate a random sample $X_{(1)} < X_{(2)} < \dots < X_{(21)}$, where $X_1, \dots, X_{21}$ are i.i.d. exponential random variable with mean 1. Repeat this process 1000 times, and based on these samples estimate $A = E(X_{(11)})$ and $B = V(X_{(11)})$. [3+4=7]

Answers: The values of

$$A = 0.7231 \quad [\text{Around } 0.7]$$

$$B = 0.0539 \quad [\text{Around } 0.05]$$

2. Generate $X_1, \dots, X_{1000}$ from $f(x)$, where

$$f(x) = \begin{cases} c^{-1}x^2e^{-4x^2} & \text{if } x > 0 \\ 0 & \text{if } x \leq 0, \end{cases}$$

where $c = \int_0^\infty x^2 e^{-4x^2} dx$. Report $A = X_{(900)}$ and $B = X_{(950)}$. [4+4=8]

Answers: The values of

$$A = 0.88 \quad [\text{Approx } 0.9]$$

$$B = 0.97 \quad [\text{Approx } 1]$$